

Performance Testing

Date	23 June 2026
Team ID	SWTID-2026-1377
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	/analyze endpoint (Voice-Based Concept Understanding Analyser)
Test Environment	Local (Flask Development Server)
Test Date	20 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits prediction form once	1	10	Analysis report generated successfully without errors
2	Multiple users submit predictions concurrently (normal load)	20	60	All requests processed successfully with acceptable response time
3	Higher concurrent load (peak usage simulation)	50	120	System remains stable and generates analysis reports correctly

4	Sustained load over longer duration	30	300	No application crashes or memory leaks; consistent response time maintained
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 3 seconds	1.5 seconds	Pass	NLP processing completed quickly for normal-length responses.
2	Response Time (Max)	< 6 seconds	3.8 seconds	Pass	Maximum observed during peak concurrent requests
3	Throughput (Req/sec)	—	18 req/sec	Pass	Flask application handled concurrent concept evaluations efficiently
4	Error Rate	< 1%	0%	Pass	No failed requests observed during testing.
5	CPU Utilization	< 80%	48%	Pass	Moderate CPU usage during NLP processing.
6	Memory Utilization	< 80%	42%	Pass	Memory consumption remained stable throughout testing.

Step 4: Observations & Analysis

Key Findings:

The **Voice-Based Concept Understanding Analyser** performed efficiently under both normal and moderate concurrent loads. The Flask backend successfully processed speech-to-text conversion, NLP preprocessing, semantic similarity analysis, and personalized feedback generation within acceptable response times. No request failures or application crashes were observed during testing..

Bottlenecks Identified:

The Flask development server is intended primarily for development and testing. Under significantly higher traffic (hundreds of concurrent users), response times may increase due to the single-process development server. NLP processing may also become more computationally intensive as the knowledge base grows.

Optimization Steps Taken:

- ☐ Optimized NLP preprocessing pipeline.
- ☐ Reused loaded knowledge base to avoid repeated file reads.
- ☐ Reduced unnecessary computations during similarity analysis.
- ☐ Structured backend modules for efficient execution